PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	The Effect of Sedation with Inhaled Anesthetics on Cognitive and Psychiatric Outcomes in Critically Ill Adults: A Systematic Review Protocol
AUTHORS	Cuninghame, Sean; gorsky, kevin; Francoeur, Conall; Withington, Davinia; Burry, Lisa; Jerath, Angela; Slessarev, Marat

VERSION 1 – REVIEW

REVIEWER	Flinspach, Armin Hospital of the Goethe University Frankfurt, Department of Anaesthesiology, Intensive Care Medicine and Pain Therapy
REVIEW RETURNED	29-Jun-2021

GENERAL COMMENTS	Congratulations on the development and submission of this study protocol for the planned, very interesting review. I would like to make a few comments regarding the manuscript. It remains largely open how the identification of relevant keywords is to be done together with the clinical librarian. Since this is a highly relevant point for locating relevant literature in the mentioned databases for data collection, a more detailed explanation would be desirable. For example, whether if so how a structured separation should be designed with regard to surgical and intensive care applications It remains unclear why the review excludes children from the collective. Especially considering the relatively frequent use in therapy-refractory status asthmatics in PICU, it should be explained why this supposedly interesting patient group is not included. At various points in the submitted manuscript, the authors provide a planned breakdown according to the time of application for volatile sedatives. However, it is not clear from the inclusion/exclusion parameters which minimum time interval was used for the definition of ICU sedation. Here it would be useful to define a distinction from prolonged intraoperative use, especially with regard to very short intervals in the range of a few hours. It would be helpful to reveal the qualifications of the independent reviewers. Are they specialists in anaesthesia with experience in the use of volatiles in intensive care environment or specialists in psychiatry or neurology?

REVIEWER	Toft, Palle Odense University Hospital, Anaesthesia and Intensive Care
REVIEW RETURNED	30-Jun-2021

GENERAL COMMENTS	The authors present a plan for a systematic review that will assess cognitive and psychiatric outcome with inhaled sedation compared

	to intravenous sedation.Three databases will be used. There is a plan for inclusion and exclusion of studies..The PRISMA system will be used.The authors expect few RCT and the references for this review consists mainly of other reviews or meta-analyses. The search strategy is rather weakly described (with the aid of our libranian).It might be possibly to present a more detailed preliminary search strategy.. I agree that intravenous sedation contributes to delirium.To claim that intravenous sedation is associated with cognitive dysfunction is more controversial.In this context the authors mention 4 references. Three of which are reviews from 2012-2019.and the last a cohorte sudy by Girard et al..Girard et al found that delirium in sedated patients (sedative associated delirium) predicted a worse cognitive outcome.In the reference list a newly published RCT is missing.In this RCT intravenous sedation did not result in long lasting cognitive dysfunction..(Crit Care Med 2020;48:1790-98)

REVIEWER	Suleiman, Aiman Beth Israel Deaconess Medical Center
REVIEW RETURNED	13-Jul-2021

GENERAL COMMENTS	Dear editor, Thank you for the opportunity to review this manuscript. The article is well-written and addresses an important subject: Notes to authors: 3 major points: 1- Cognitive impairment is a wide umbrella shadowing many diseases and diagnoses, there should be a scientific ground for your inclusion of the spectrum of diseases (you can't just give examples), I suggest to use ICD-9 or 10 definitions, or a 'clinically plausible literature derived definitions' (although I find the late difficult to address). 2- Your inclusion/exclusion criteria should hit 2 major points: a. Patients who received any kind of IV sedation before, during, or after inhalational sedation, are they included? (be carefull that most of the studies you will find are observational, many will allow certain conditions of IV sedation) b. Patients with previous cognitive impairments or on psychiatric medications (that's why definitions are crucial) 3- Inhalational sedation for ARDS patients during COVID-19 pandemic seems plausible, but there are many reviews that pointed the pros and cons of this (read Suleiman et al. Sedating Mechanically Ventilated COVID-19 Patients with Volatile Anesthetics: Insights on the Last-Minute Potential Weapons. Sci. Pharm. 2021, 89, 6.). This is to say that your introduction points towards a 'flawless' approach to sedation, while in fact, it is a two-sided coin with many limitations from practical point of view. Thank you.

REVIEWER	Widrich, Jason University of Florida College of Medicine, Anesthesiology
REVIEW RETURNED	20-Jul-2021

GENERAL COMMENTS	This study protocol is an interesting and important question. Without even considering anesthetics, there have been a number of different measures used through the years to identify delirium,

	psychiatric/personality changes, longer term impairment post operative cognitive dysfunction, and dementia that make this a difficult topic to study for any one intervention for a meaningful statistical analysis.. From reading your study design, hopefully you will find enough papers on anesthetics to be able to parse out those categories of dysfunction and not combine them Thank you.

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1
Dr. Armin Flinspach, Hospital of the Goethe University Frankfurt
Comments to the Author:
Congratulations on the development and submission of this study protocol for the planned, very interesting review.
I would like to make a few comments regarding the manuscript.

Comment 1: It remains largely open how the identification of relevant keywords is to be done together with the clinical librarian. Since this is a highly relevant point for locating relevant literature in the mentioned databases for data collection, a more detailed explanation would be desirable. For example, whether if so, how a structured separation should be designed with regard to surgical and intensive care applications.

Response 1: Thank you Dr. Flinspac, we believe the edits made based on your request for further clarification have resulted in a more clear, more comprehensive manuscript. Key edits can be found under the Search Strategy on page 8, paragraph 2. We are a multidisciplinary team of various clinical and methodological backgrounds and we have worked with our certified clinical librarian to develop the process and identify keywords (i.e., MeSH terms) to be used in our search. The process involved an initial search of relevant words to identify keywords as well as drawing on our previous knowledge and systematic reviews on this subject matter (Honarmand et al., 2020). To increase transparency, we have attached our finalized search strategy Supplemental Information. We have also acknowledged and included additional statements on p/span>age 8 of our Search Strategy heading to ensure we are following the updated PRISMA-S Checklist (released in 2021, designed specifically for reporting on the methods of the literature search; Rethlefson et al., 2021). We hope this brings clarity to our search strategy as well as to how keywords were identified and will be used to locate relevant studies.

Comment 2: It remains unclear why the review excludes children from the collective. Especially considering the relatively frequent use in therapy-refractory status asthmatics in PICU, it should be explained why this supposedly interesting patient group is not included.

Response 2: Thank you for raising this important point. We agree that assessment of these outcomes in children is of interest to the pediatric critical care community. While designing the protocol we discussed if children should be included in our review. Given our previous experience examining cognitive outcomes in adults (Honarmand et al., 2020), we found that children are assessed for cognitive outcomes using a variety of other tools not common or applicable to adults. Furthermore, assessment of their cognition across a range of developmental stages makes it difficult to draw meaningful comparisons to adult populations. Therefore, in this review, we will focus on the adult population. Authors from within our group however, in collaboration with the Canadian Critical Care Trials Group, aim to pursue study of the pediatric population in the future.

Comment 3: At various points in the submitted manuscript, the authors provide a planned breakdown according to the time of application for volatile sedatives. However, it is not clear from the

inclusion/exclusion parameters which minimum time interval was used for the definition of ICU sedation. Here it would be useful to define a distinction from prolonged intraoperative use, especially with regard to very short intervals in the range of a few hours.

Response 3: Thank you for bringing this issue to our attention. We have made it more clear in the protocol (page 6, paragraph 4) that patients may be included as long as they received inhaled sedatives for any period of time in the ICU, as long as they satisfy the other inclusion criteria (examined for cognitive outcomes, etc.). We purposefully chose not to limit the duration of ICU exposure, as this relationship is currently not well described and may impact cognitive outcomes (e.g.. longer exposure resulting in worse cognitive outcomes)

We have clarified this in line 132-150 of the revised protocol to read, " Patients who received inhaled sedatives for any period of time during their admission to an intensive care unit. Treatment could be initiated in the ICU, the operating room, or the emergency department."

Comment 4:It would be helpful to reveal the qualifications of the independent reviewers. Are they specialists in anaesthesia with experience in the use of volatiles in intensive care environment or specialists in psychiatry or neurology?

Response 4: Our team is made up of a multi-disciplinary team including trainees with background training in internal medicine and anesthesia (SC & KG). Our senior researchers on this team include pediatric and adult attending intensivists [base specialty training in internal medicine (MS), pediatrics (CF), and anesthesia (DW, AJ)] with additional graduate training in research methodology (MS, AJ), and demonstrated research productivity in the area of neurocognitive outcomes (CF, DW, LB, AJ, MS). Furthermore, we have collaborated with our Clinical Pharmacy Specialist who is a career Clinical Scientist in the Faculty of Pharmacy with extensive research experience on drug safety in the ICU, systematic reviews, and trial design (LB). If the Editors feel strongly that this information be included as part of the body of the text, we would be happy to include a statement on this as part of the Author's Contribution heading.

Reviewer: 2
Dr. Palle Toft, Odense University Hospital
Comments to the Author:
Comment 1: The authors present a plan for a systematic review that will assess cognitive and psychiatric outcome with inhaled sedation compared to intravenous sedation.Three databases will be used. There is a plan for inclusion and exclusion of studies..The PRISMA system will be used.The authors expect few RCT and the references for this review consists mainly of other reviews or meta-analyses.
The search strategy is rather weakly described ("with the aid of our libranian"). It might be possibly to present a more detailed preliminary search strategy..

Response 1: Thank you for the suggestion to be more thorough in our description of the search strategy. We have expanded the description of our search strategy under the "Search Strategy" heading on page 8. This is a similar suggestion to that raised by Reviewer #1 regarding elaborating on keyword choice (see above Comment #1 for a related response).

Comment 2: I agree that intravenous sedation contributes to delirium.To claim that intravenous sedation is associated with cognitive dysfunction is more controversial.In this context the authors mention 4 references. Three of which are reviews from 2012-2019.and the last a cohorte sudy by Girard et al..Girard et al found that delirium in sedated patients (sedative associated delirium) predicted a worse cognitive outcome.In the reference list a newly published RCT is missing. In this

RCT intravenous sedation did not result in long lasting cognitive dysfunction. (Crit Care Med 2020;48:1790-98).

Response 2: Thank you for highlighting this issue. We have amended the manuscript to make it clear that there is conflicting evidence surrounding the use of IV sedation and long-term cognitive outcomes. Specifically, we have included the reference (Crit Care Med 2020;48:1790-98 PubMed) as part of our attempt to clarify. The updated text that reflects these changes can be found on lines 90-102.

Reviewer: 3
Dr. Aiman Suleiman, The University of Jordan
Comments to the Author:
Dear editor,
Thank you for the opportunity to review this manuscript.
The article is well-written and addresses an important subject:
Notes to authors:

3 major points:
Comment 1: Cognitive impairment is a wide umbrella shadowing many diseases and diagnoses, there should be a scientific ground for your inclusion of the spectrum of diseases (you can't just give examples), I suggest to use ICD-9 or 10 definitions, or a 'clinically plausible literature derived definitions' (although I find the late difficult to address).

Response 1: Thank you for raising this thoughtful point. We absolutely agree that the term 'Cognitive Impairment' is broad, and often used interchangeably with many specific diagnoses. Unfortunately, based on our experience, this type of classification (using specific ICD-9 or 10 definitions) is often lacking in studies involving critically ill patients. In fact, the need for specific definitions and classifications of impairments is precisely why we think this review will be valuable to the critical care community in order to highlight this deficiency. By using a broad MeSH keyword such as "Cognitive Impairment", we want to ensure we capture all possible definitions used by authors in this area. The MeSH terms included "Cognitive Impairment" and also includes these broad definitions that may be used interchangeably with it (e.g., post-operative delirium versus confusion versus sepsis-associated encephalopathy, etc.) used in various critical care studies, where, in our experience, there is a lot of heterogeneity of the terminology used (Honarmand et al., 2020). It is precisely our goal to help elucidate the many cognitive diseases, processes, and diagnoses that critically ill adults may present with, so by using a broad definition, we expect to capture and characterize these specific disease processes.

Comment 2: - Your inclusion/exclusion criteria should hit 2 major points:
a. Patients who received any kind of IV sedation before, during, or after inhalational sedation, are they included? (be careful that most of the studies you will find are observational, many will allow certain conditions of IV sedation)
b. Patients with previous cognitive impairments or on psychiatric medications (that's why definitions are crucial)

Response 2: We have included patients who received adjunctive IV sedation as part of their sedation management during their ICU stay and this will be included, where applicable, in our data extraction. We have added this clarification to lines 162-163 of the protocol to read, "Patients who received adjunctive intravenous sedation (e.g., concomitant benzodiazepiens or opioids) will be included as long as they satisfy the above inclusion criteria." Similarly, we have updated the "Population Characteristics" subgroup of our described data extraction heading to include the following on lines 212-213, "Previous diagnosis of cognitive impairment, PTSD, Anxiety, or Depression, previous

antipsychotic use." to ensure this is discussed, should this information be included in the results from our included studies.

Comment 3: - Inhalational sedation for ARDS patients during COVID-19 pandemic seems plausible, but there are many reviews that pointed the pros and cons of this (read Suleiman et al. Sedating Mechanically Ventilated COVID-19 Patients with Volatile Anesthetics: Insights on the Last-Minute Potential Weapons. Sci. Pharm. 2021, 89, 6.). This is to say that your introduction points towards a 'flawless' approach to sedation, while in fact, it is a two-sided coin with many limitations from practical point of view.

Response 3: Thank you for bringing this to our attention. We agree that there are practical difficulties in establishing inhaled anesthetics in the ICU. In addition, we recognize that they do not come without some potential adverse effects. We have included points on this in our introduction to ensure we are not biasing inhaled anesthetics as a flawless approach to sedation. We have included your 2021 review as well which highlights these important considerations. The changes can be found on page 4, paragraph 1 of the revised manuscript.

Reviewer: 4
Dr. Jason Widrich, University of Florida College of Medicine
Comments to the Author:

Comment 1: This study protocol is an interesting and important question. Without even considering anesthetics, there have been a number of different measures used through the years to identify delirium, psychiatric/personality changes, longer term impairment post operative cognitive dysfunction, and dementia that make this a difficult topic to study for any one intervention for a meaningful statistical analysis.. From reading your study design, hopefully you will find enough papers on anesthetics to be able to parse out those categories of dysfunction and not combine them. Thank you.

Response 1: Thank you for raising this important point. We agree, with the multiple interventions imposed during an ICU stay, it can be difficult to determine if any sole intervention is the cause of a particular outcome. We believe that performing this review will help highlight the fact that we know little about how sedatives in the ICU affect outcomes and will serve as a call to help investigate their effects further.

Reviewer: 1
Competing interests of Reviewer: The Reviewer received speaker fees from P.J. Dahlhausen & Co. GmbH, Colone, Germany and financial support through the Sedana Medical Research Grant (Sedana Medical AB, Danderyd, Sweden).

Reviewer: 2
Competing interests of Reviewer: No competing interests

Reviewer: 3
Competing interests of Reviewer: None

Reviewer: 4
Competing interests of Reviewer: none

REFERENCES:

1. Honarmand, K., Lalli, R. S., Priestap, F., et al. Natural history of cognitive impairment in critical illness survivors: A systematic review. American Journal of Respiratory and Critical Care Medicine, 2020; 202(2), 193-201.
2. Rethlefsen, M. L., Kirtley, S., Waffenschmidt, S., Ayala, A. P., Moher, D., Page, M. J., & Koffel, J. B. (2021). PRISMA-S: an extension to the PRISMA statement for reporting literature searches in systematic reviews. Systematic reviews, 10(1), 1-19.

VERSION 2 – REVIEW

REVIEWER	Flinspach, Armin Hospital of the Goethe University Frankfurt, Department of Anaesthesiology, Intensive Care Medicine and Pain Therapy
REVIEW RETURNED	18-Oct-2021

GENERAL COMMENTS	Thank you very much for the further explanations of your work.